Dynamic Portfolio for Personalized CPAP Treatment: Adaptive Digital Biomarker-Driven Strategies Across the OSA Care Pathway

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***Abstract*—Obstructive Sleep Apnea (OSA) management struggles from static retrospective adherence models which build their assessment on rigid compliance benchmarks. The study introduces an Artificial Intelligence-based framework for CPAP treatment adaptation which combines real-time biosensor data and personalized treatment stages. The system categorized patients based on therapy dynamics through unsupervised learning analysis into four stages new, adherent, attempting, and non-adherent while considering CPAP usage data and clinical histories and Monitoring survey results. The adopted monitoring system tracks a selection of digital biomarkers (e.g., HRV, ODI, RVO) in real-time to identify initial signs of therapy deviation as well as changes in physiological responses. It enables the risk scoring system to set off weekly risk evaluations and adjust cluster groupings and activate custom-made intervention plans such as digital prompts and medical survey requests alongside physician alert actions. Continuous physician feedback connections enable healthcare providers to check and modify or restructure interventions to ensure clinical accuracy and security. The system provides dynamic adjustment of biomarker approaches alongside intervention force levels which depend on patients' current therapy state and clinical complexity. The proposed framework displays promising results to detect patients at risk for dropping out and to optimize their therapy according to preliminary retrospective data analysis. The developed work provides a scalable and explainable clinical framework that will support future tests and deployments for biomarker-driven real-time CPAP personalization methods throughout the OSA care pathway.**

Keywords— Obstructive Sleep Apnea (OSA), Continuous Positive Airway Pressure (CPAP), CPAP Adherence, Personalized Care, Biometric Feedback, Digital Biomarkers, Adaptive Interventions, AI-driven Personalization, Risk Assessment, Sleep Therapy Optimization.

I. INTRODUCTION

Obstructive Sleep Apnea (OSA) as a common sleep-related breathing disorder which produces upper airway blockages while sleeping those results in hypoxemia, small arousals and destruction of regular sleep patterns. OSA impacts up to 38% and as low as 9% of the adult population worldwide while it causes extensive cardiovascular harm because of prolonged sympathetic overactivation and systemic inflammation [1], [2]. Continuous Positive Airway Pressure functions as the main evidence-based therapy for treating moderate to severe Obstructive Sleep Apnea by delivering continuous airflow to stop airway collapse. The medical benefits of CPAP do not match its actual usage success rates among patients. up to 60% of patients discontinue CPAP within the first year, primarily due to discomfort or low perceived benefit [3], [1]. Patient therapy dropout is not due to lack of willpower but represents a missed chance for clinicians to detect disengagement indicators earlier [4]. This failure points to reactive and static monitoring approaches as the source of the problem [4]. The current approach relying on retrospective metrics such as 4-hour CPAP minimum use [5]. This traditional method simplifies patient needs while hiding potential biological and behavioral issues. The standard clinical diagnostic approach through polysomnography (PSG) maintains its status as the most reliable method yet remains strenuous to operate in both ongoing and large-scale real-time monitoring because of its intrusive nature [6], [7]. Today's wearable technology and remote monitoring systems present opportunities to monitor OSA in real-time and understand its evolution as well as treatment responses. Real-time physiological assessment for autonomic stress and respiratory instability is possible through digital biomarkers such as HRV, ODI, and RVO. The sensors used to detect respiratory signals through thoracic bands and ECG monitors and oximetry probes present challenges because they cause discomfort for patients who need extended home care monitoring [8]. Clinical use of biomarkers is limited due to interpretation and infrastructure challenges. Our study develops an intelligent AI-based framework which uses biomarkers to deliver individually tailored CPAP administration. The system evaluates patient therapy phases that evolve between new and adherent and attempting and non-adherent then uses continuous physiological and

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behavioral signal tracking to make decisions. Real-time risk scores emerge from inputs which control dynamic intervention plan changes alongside physician activation protocols. The system delivers intervention frequency based on patient complexity ratings to provide higher-frequency monitoring and intervention loops to patients with unstable biometrics along with comorbidities [4], [9]. The physician validation process is built directly into the system structure because it functions to support intervention changes through threshold monitoring [9], [10]. The departure of patients from therapy is redefined from therapeutic failure to early warning sign of detection failure which demonstrates the importance of proactive risk monitoring systems [4]. Personalized treatment should adjust dynamically throughout different stages of treatment because it needs to correspond with patient comorbidities and therapy patterns together with risk changes [11]. The applications of AI in OSA management achieved segmentation of patient trajectories using clustering models which incorporated k-means, Mean Shift and Hidden Markov Models [3], [12]. Most models operate with set baseline features while maintaining rigidity and lack abilities to track treatment phase changes and patient-reported disease experiences and comorbidity patterns. The absence of dynamic treatment logic improvement has resulted in underwhelming clinical use rates and poor customization outcomes [13], [14]. The presented research outlines an AI-supported modular system for respiratory therapy adjustment that joins patient group categorization with HRV analysis and ODI monitoring and SpO_2 tracking along with biological signal tracking and dynamic medical protocol integration. The framework is designed to adapt across therapy phases and patient profiles, accounting for comorbid conditions such as cardiovascular disease, obesity, or depression, by modulating intervention intensity and monitoring strategies based on both physiological trends and behavioral feedback. Clinician collaboration stays essential since it helps track complicated biomarker signals to maintain treatment appropriateness and explanation. This system possesses capabilities to expand across different scales and maintain trackable documentation and adjust to healthcare needs of changing patients and comply with medical standards and regulatory requirements.

II. State of Art

A. Scope and Depth of Literature Review

To develop a rigorous and evidence-grounded framework for CPAP personalization, we conducted a structured review of over 3,000 academic and recent clinical sources indexed in Web of Science between 2022 and 2025. Our review covered 492 studies on CPAP adherence, 484 on therapy discontinuation, and 487 on behavioral and psychosocial adherence barriers. In addition, we examined 64 articles focusing on digital biomarkers, 51 on HRV and sleep-stage transitions, and 123 articles on CPAP interventions using physiological signals. Wearable biosensors such as smartwatches, oximeters, and CPAP-integrated sensors enable real-time, passive measurement of respiratory signals (AHI, ODI), cardiac autonomic markers (HRV), and behavioral feedback loops [9]. Studies confirm that combining these tools with behavioral nudges or personalized education improves adherence and cluster detection [12].

B. Gaps in Traditional CPAP Adherence Models

The standard therapy for treating moderate to severe Obstructive Sleep Apnea (OSA) with Continuous Positive Airway Pressure (CPAP) exhibits low real-life patient adherence levels that result in 40–50% treatment discontinuation rates during initial treatment year usage [13], [11]. Current adherence systems depend on strict definition standards of at least four hours of night-time usage which CPAP devices monitor after the therapy has finished [5]. Phase-based monitoring absence leads health providers to treat patients at various treatment stages the same manner since they do not consider vital differences in adherence histories and risk profiles [15]. Behavioral studies underscore the importance of psychosocial factors such as mask discomfort, anxiety, and perceived stigma as primary drivers of early dropout [16]. Recent reviews establish that therapy responds poorly to disruption since routine feedback mechanisms do not exist [5].

C. Underutilization and Fragmented Use of Digital Biomarkers

Modern wearable technology now allows for time sensitive physiological measurement of significant data points for CPAP therapy operations. The combination of digital biomarkers includes heart rate variability (HRV) and oxygen desaturation index (ODI) alongside apnea-hypopnea index (AHI) and respiratory effort variability which gives physicians early diagnostic information about treatment effectiveness and remaining illness effects or autonomic system dysfunction [8]. However, current CPAP adherence models neither incorporate these signals nor interpret their patterns in relation to individual treatment stages or comorbidity burden. For example, an adherent patient with stable metrics may not require daily tracking, while a newly relapsed patient might benefit from enhanced HRV and ODI monitoring [11]. Clinical workflows often assess these signals retroactively or uniformly across patients, without consideration for therapy stage or individual risk. This lack of phase-aware adaptation limits personalization and overwhelms physicians with data that lacks context-specific relevance. By applying this generalized technique clinicians face an elevated challenge of interpreting collected data without clear proof of its meaning or specific clinical purpose.

D. Limitations of Static Clustering and Non-Adaptive Risk Stratification

Several researchers have examined machine learning methods to cluster OSA patient data for understanding CPAP patterns and non-compliance risks and syndromic differences between patient groups [17], [9]. AI-driven clustering has been widely used to segment OSA patients based on usage, biometrics, and survey data. Clustering methods such as K-means [12], hierarchical clustering, and Mean Shift have shown effectiveness in identifying phenotypic subtypes with varying risk profiles and adherence behavior [18], [19]. Such static models are particularly problematic for mid-stage or attempting patients those restarting therapy after dropout who require customized monitoring and support structures. These models approach clustering procedures as a single processing step, yet they do not adapt across time. After initial cluster assignment the patient stays in their assigned group while their physiological condition and behavioral data points may change through the course of treatment [12]. Such problems are particularly severe for patients in the intermediate stages of treatment who must readopt their therapy following dropout or switch to self-managed care from hospital supervision. The analysis of clustering data fails to integrate external information from behavioral qualifiers like mask fit quality or

treatment comprehension and lifestyle adjustments that research demonstrates influence patient adherence [13]. Recent research demonstrates the need for dynamic clustering models which make recurring patient group adjustments through updates as new sensor data or clinical surveys and physician annotations become accessible [12], [20]. Better intervention grouping is possible while phase-dependent care approaches and clinical understanding of model interpretation become more transparent through this framework.

E. Toward a Phase-Aware, Feedback-Driven CPAP Treatment Portfolio

Emerging literature and expert consensus underscore the need for modular, adaptable treatment pathways what we define as a "treatment portfolio" to guide CPAP personalization across therapy phases [21], [22]. Treatment portfolios differ from fixed adherence models because they allow dynamic changes in intervention approaches through biomarker deviations and engagement signals and therapy-phase progress. Portfolio strategies must distinguish between:

- **Initial therapy setup:** where patients need onboarding support and early symptom calibration [23].
- **Mid-therapy optimization:** The system performs ongoing measurements of physical indicators and behavioral signs (fatigue, mood and treatment engagement) to modify risk parameters and therapy participation strategies automatically [24].
- **Late-stage or dropout recovery:** Demands more intensive tracking, decision-tree-based interventions, and escalated outreach (e.g., physician alerts, technical follow-up) [25].

Multiple studies prove that real-time healthcare systems with event-based alerting mechanisms partnered with adaptive AI feedback systems and interactive medical reporting elements help lower therapeutic inertia and enhance long-term adherence rates in OSA patients along with patients in chronic care programs [5], [26]. These systems demonstrate the actual nature of treatment paths because outcomes follow unpredictable directions while therapy needs growth continuously.

F. Medical and Technological Imperatives for Integrated Frameworks

CPAP personalization frameworks involve combining medical logic approaches with computational designing principles as part of their systems development framework. A clinical implementation of the approach needs to operate within physician clinical processes and enable decision-tree applications together with contextual warning systems to trigger prompt interventions. The model requires combination of clustering methods with digital biomarkers integration and real-time monitoring and explanation capabilities for implementation success [4]. Recent studies show that models integrating machine learning with structured medical surveys (e.g., PSQI, ESS, BDI) and CPAP usage data offer higher personalization, better adherence, and increased physician trust [12]. Frameworks that continuously evaluated feedback and adjust clustering and interventions show a 2.4× improvement in adherence and better sleep quality outcomes [12]. The proposed framework operates between phases while processing diverse data streams through AI models which maintain clinical usefulness.

III. METHODS

This research develops a completely dynamic AI-based adherence model that improves CPAP therapy success rates for OSA patients. The suggested system addresses limitations of static adherence monitoring through its blend of biometric tracking with advanced machine learning classification adaptive interventions, and real-time physician input. The system enables instant personalized and responsive patient care through deep understanding of patient physiological status and behavioral feedback. This methodology is rooted in evidence from multiple clinical studies and informed by domain-specific research [27], [9].

A. Patient Selection and Initial Clustering

The study begins by classifying patients into distinct groups based on their CPAP adherence levels with new patients having no recorded usage and adherent patients (≥4 hours of nightly usage), attempting patients (2–4 hours nightly), and non-adherent patients (<2 hours nightly). The method allows researchers to deliver personalized treatment interventions which specifically target the phase of patient therapy [22]. Clustering uses extensive retrospective and real-time multidimensional datasets that implement unsupervised machine learning algorithms, primarily Mean Shift, k-means clustering. This comprehensive clustering considers: the essential demographic data used for predicting therapy adherence includes age together with gender and BMI and psychosocial factors [28]. The detailed metrics obtained from CPAP devices include usage duration and mask leakage variability together with residual AHI and pressure settings [15]. Monitoring surveys contain a full set of behavioral data along with adherence patterns, therapy interruption records, preferred mask styles, activity measurements and cognitive assessment results and mood evaluations [29]. Medical survey responses that use validated measurement tools including PSQI, ISI, ESS, FSS and SF-36, BDI lead to improved predictive clustering performance [12]. During the initial 7–14 days of new patient monitoring is prioritized. The initial data helps identify early risk indicators, which allow medical teams to identify patients requiring fast medical intervention.

B. Risk Evaluation and Stratification

The risk assessment combines several behavioral and physiological indicators which are computed on a weekly basis.

- Usage trend (7-day average)
- Leak level and stability
- Residual AHI fluctuation
- The risk assessment combines survey data points from ESS increase and fatigue complaints.
- Intervention efficacy (usage before/after response)

Machine learning is used to dynamically update risk thresholds. A system of detection classifications arranges patients into these four distinct groups based on risk level from low to critical. The risk scoring process for new patients takes diagnostic severity along with early physiological signals to produce their starting score. The chosen modeling approach follows best practices of adaptive care and effectively matches historical dropout patterns from prior research [30], [8].

C. *Digital Biomarker Selection and Evaluation*

Digital Biomarkers select from dynamic combinations based on therapy phase together with risk classification and patient response history. The assignment is as follows:

- Health risk variables and oxygen saturation will be measured initially and RVO will be monitored throughout 7–14 days for new patients
- Adherent patients: Monthly HRV and AHI trend monitoring
- The assessment of Attempting patients consists of weekly HRV analysis together with ODI evaluation and symptom survey collection
- Non-adherent patients: Daily HRV, SpO_2, and ODI tracking

The choice of wearable devices Hexoskin [31] and Somno-Art [32] contributes to validate multi-channel physiological data without causing significant patient disruption. The inclusion and monitoring frequency of biomarkers depends on signal quality together with dropout prediction accuracy coupled with changing patient responses [5]. This modular, feedback-driven approach ensures that monitoring remains relevant, targeted, and resource-aware.

D. *Adaptive Intervention Implementation*

The framework assigns tailored interventions based on individual patient adherence position and risk assessment results and behavioral reactions. Real-time decision-based algorithms along with machine learning mechanisms operate within the intervention module to deliver on-time support. Intervention design includes:

- **New patients:** we provide education and devices with motivational content in combination with survey-based follow-up.
- **Adherent patients**: Low-intensity monitoring, gamification, and engagement reinforcement
- **Attempting patients:** The interventions consist of behavioral nudges together with real-time alerts and weekly coaching prompts.
- **Non-adherent patients**: High-frequency alerts, digital coaching, physician intervention

By using machine learning processes biometric and survey inputs to enable just-in-time interventions. The adaptive feedback loops reduce therapy disruptions by helping patients maintain their treatments which results in better clinical results and treatment adherence [9].

E. *Cluster Refinement and Continuous Learning*

The healthcare team evaluates all patient clusters and risk assessment models each 4–6 weeks or when major behavioral events occur such as treatment abandonment or non-responsiveness to care measures. The model retraining process includes new variables (post-intervention HRV and biomarker anomalies plus survey deltas) while validity tests use Silhouette Score and intervention conversion rate. By adapting the approach research scientists minimize overfitting effects but achieve better personalization results. The proposed system enables a phase transition process which changes attempted users to adherents after evaluating new digital biomarkers together with usage data [11].

F. *Physician Evaluation and Feedback Loop*

Real-time patient-reported outcomes along with dynamic biometric signals input to the proposed machine learning system leads to personalized intervention strategies that produce specific context-based responses. The adaptive feedback loops help prevent interrupted ways from helping health professionals detect disengagement patterns to deliver prompt assistance which leads to enhanced therapeutic effects while promoting longer CPAP usage. The detailed diagram in Fig.1 represents the entire system interactions which show the end-to-end process from physician-led diagnostic testing to CPAP customization steps and AI Framework followed by biological parameter survey and escalation procedures.

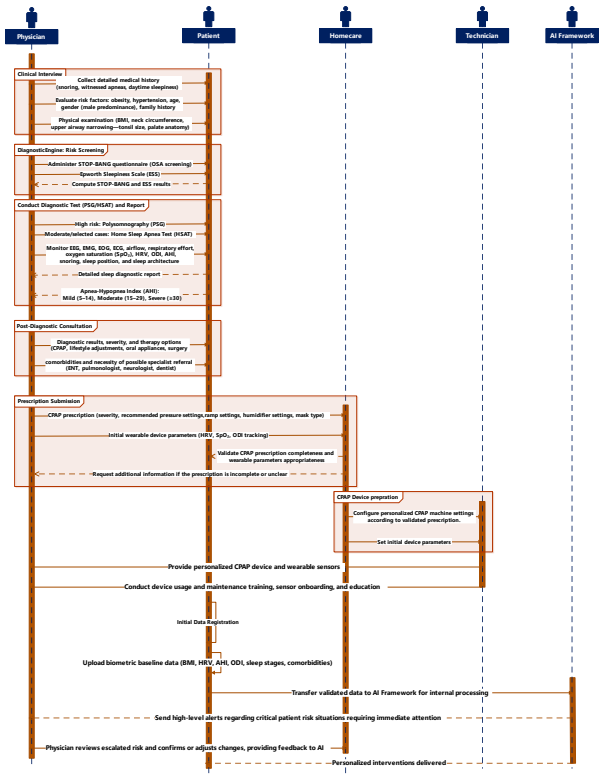

Fig 1: Initial Diagnostic Workflow and CPAP Setup Pathway with Physician Oversight

The design components unite into an integrated framework which provides clinical evidence-based and technologically capable patient adherence solution. The decision-making process between AI and healthcare professionals creates an architecture where AI works as an assistant to support clinical practice. Patient feedback, evolving risk scores, and biometric updates drive a dynamic adherence system guided by physicians. To maintain pathway coherence, a built-in consistency ensures alignment between AI suggestions and clinical standards, automatically flagging deviations for review. This safeguards treatment continuity and supports transparent, phase-consistent care decisions. The framework exists as a remedy for known gaps in static monitoring systems and creates pathways to conduct ongoing clinical testing that will lead to large-scale implementation.

IV. IMPLEMENTATION

Under real-world implementation this section defines an explainable phase-based process which leverages data to assess therapy risk levels while selecting digital biomarker

strategies and personalizing interventions Fig.2. The pending deployment of this framework incorporates its system structure along with the best practices and discussion-based strategic logic which was developed with supervisory input.

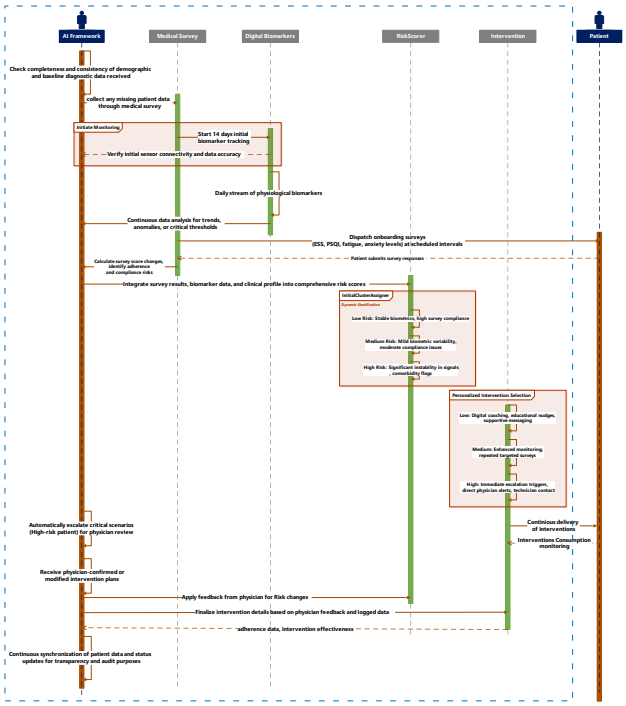

Fig 2: Interaction between patients and proposed AI system for intervention delivery

A. *Data Collection and Preprocessing Framework*

The research methods followed methods from previous studies [12] by collecting extensive data from Philips and ResMed and Sefam and Lowenstein CPAP devices along with detailed patient medical files and annotated risk factors and treatment histories and patients surveys using validated monitoring questionnaires. Multiple patient engagement channels served to collect the data including mobile applications and teleconsultations and SMS outreach and in-person follow-ups according to [5]. The data preprocessing included multiple standard methods which used SimpleImputer to address missing data then followed with LabelEncoder for label transformation and StandardScaler for numerical standardization and ColumnTransformer to process multiple feature types simultaneously. The established data engineering pipeline established a stable framework for risk-informed clustering as well as adaptive intervention mapping through which downstream biomarkers could be integrated.

B. *Initial Clustering Iteration: CPAP Usage Profiles*

Analysis during the initial clustering phase included CPAP usage data such as AHI, LeaksLargePercentage, and estimated nightly use. The mean shift clustering algorithm recognized clusters of any shape. Mean Shift clustering identified nine data-driven clusters with Silhouette Score achieved 0.37 along with Davies-Bouldin Index at 0.76. Patient group patterns revealed distinct adherence challenges: low usage, excessive leaks, and high residual AHI levels, each requiring tailored interventions. The analysis yielded specific biomarker profiles. These patterns guided early clinical biomarker monitoring strategies. The analysis patterns in this study match precisely with information in [28].

C. *Intermediate Clustering: Risk Factor and Demographic Integration*

In the next iteration, the clustering model incorporated patient-level demographics (birth date, gender, and postal code) together with mask types along with their risk factors. These static elements strengthened risk prediction when biometric signals were unavailable. ANOVA analysis confirmed that three key variables among five initial cluster groups were device type alongside leak percentage and average usage duration. This refinement yielded nine more coherent clusters aligned with actionable intervention criteria.

D. *Advanced Clustering: Integrating Intervention and Monitoring Surveys*

This iteration included intervention logs with ServiceId, JobTypeCode, Status fields as well as structured survey responses which contained QuestionId, AnswerValue and ExecutionDate pairs. Mean Shift clustering produced five behavioral clusters from the expanded dataset with scores of 0.2912 Silhouette and 0.8432 Davies-Bouldin Index. High-risk patients were assigned onsite evaluations and full-length surveys (e.g., PSQI, SF-36). The survey allocation for patients with moderate risk included physician-supported assessments by combining BDI with SF-36. Low-risk patients were assigned digital self-reports using abbreviated surveys. Operational implementation of biomarker-based models into subsequent phases required the defined approach which connected survey depth to risk categories.

E. *Behavioral–Physiological Risk Scoring Framework*

A weekly composite risk score was calculated to capture behavioral and physiological instability. The risk assessment model contained four sections which included daily usage falling below an hour combined with high leak rates above 30 L/min for two or more days followed by a significant elevation of AHI events above 5 per hour with elevated ESS scores measuring more than 2 points and the lack of response from previous treatments. During model training stages the physician confirmed dropouts received maximum score points. When the scoring system generated results from 0 to 2 it indicated low risk status while moderate risk alerts were triggered between scores 3 to 4 and patients required monitoring enhancements. Results between 5 to 7 triggered automated medical procedures while results 8 or greater demanded immediate physician attention with a system-generated pathway report for patients. The report gathers current biometric variations (such as reduced HRV and increased ODI) along with behavioral signs (including usage drop patterns and increased ESS results) and previous intervention performances. The system provides physicians with information about pending tasks which include unresolved alerts as well as missed follow-ups so they can make decisions with complete understanding. This stratification structure was informed by patterns reported in [9] for its initial structure which researchers enhanced through data collected in this research[15].

F. *Digital Biomarkers Assignment and Monitoring*

The digital biomarkers system will adjust based on therapy stages and evolving risk profiles to detect early signs of non-adherence. The HRV and SpO_2 and RVO measurements will

are continuously tracked during the initial 7–14 days for new patients without CPAP history. This early window allows detection of therapy disruption using wearables like Hexoskin and Somno-Art to monitor cardiopulmonary stress [7]. For adherent patients, HRV and AHI are monitored monthly to ensure ongoing engagement. The approach for attempting patients who used the device for 2-4 hours in one night included weekly assessments that will combine HRV tracking with ODI measurements along with survey responses from ESS and PSQI evaluations referring to professional guidance about monitoring tradeoffs [25]. Non-adherent patients require daily biomarker monitoring and symptom logging for rapid intervention to identify early relapse or dropout conditions. Low-value biomarkers (e.g., unstable RVO) are deprioritized through ongoing feedback calibration. The system will adapt through sequential design of monitoring to match patient needs while keeping surveillance specific to ongoing clinical requirements and respecting each person's changing medical condition.

G. Intervention and Alert Strategy

The intervention triggers followed directly from dynamic risk scores. Middle-risk patients received empowerment and motivational guidance, and educational content delivered through digital prompts and gamified interfaces. High-risk patients received intensified coaching, adherence assistant calls, and potential pressure reassessment. Critical-risk profiles triggered automatic physician alerts incorporating biometric and behavioral history. This strategy integrates automated feedback with physician oversight for maintaining adherence and personalization [16], [13].

H. Iterative Cluster Refinement and Continuous Learning strategy

Clustering and risk scoring models are retrained every 4–6 weeks or after significant patient-related events, such as therapy dropout, non-responsiveness, or critical feedback during physician calls or digital follow-ups. Model inputs include updated usage metrics, biomarker trends, and intervention efficacy. New clustering schemas are adopted if they yield improved Silhouette Scores or enhanced intervention conversion rates. A continuous feedback loop ensures that the system adapts to evolving patient behavior, clinical status, and engagement data, maintaining therapeutic personalization and improving long-term adherence outcomes.

I. Physician Feedback Integration strategy

Physicians received information about patient treatment stages alongside evolving risk assessments combined with real-time clinical metrics such as HRV, ODI and SpO_2 and previously provided interventions through an interactive report. Physician decisions feed back into the AI engine for model refinement and learning. The framework's actual deployment merges analytics techniques with healthcare interpretability to build an adaptable system which enhances CPAP adherence awareness. AI-driven digital biomarkers monitoring is continuously adjusted under physicians' oversight. This framework enables scalable AI deployment in personalized respiratory care.

V. Conclusion

The research introduced an AI method that united predictive risk assessment with adaptive CPAP treatment portfolio protocols. The Mean Shift clustering approach was selected above static models because it automated cluster-number discovery to analyze different levels of usage and adherence behaviors. The framework accumulated multiple database types incorporating CPAP metric statistics together with leakage rates and AHI fluctuations as well as risk factors and intervention data and survey results to create a system that advanced continuously through sequential cluster transformations during each cycle. The strategy for patient intervention adjusted according to their risk status where high-risk clusters obtained enhanced support with alerts combined with biomarker analysis and medical team follow-up but low-risk clusters got monitoring through surveys in addition to feedback sessions. Through its ability to modify operational approaches the system allocated resources effectively which subsequently produced better therapeutic results. A significant advancement included the real-time process of integrating risk assessment scores with wearable biomarker selection tools for tracking purposes. Therapy-phase-specific thresholds that combined digital biomarkers such as HRV, SpO_2, ODI and RVO enabled the system to detect deviations which supported personalized care planning processes. The framework makes space for digital biomarkers to be integrated in real time for improving intervention effectiveness, but they remain outside the clustering mechanism due to their asynchronous and highly variable temporal resolution, which can introduce noise and inconsistency into the initial grouping logic. Instead, biomarkers are used post-clustering for phase-specific monitoring and intervention calibration, ensuring stability in patient segmentation while preserving flexibility in adaptive care delivery. Through this research clinical monitoring functions together with AI-controlled feedback mechanisms establish dynamic patient-oriented digital interventions instead of traditional static compliance systems. Healthcare facilities can adopt this framework because it delivers essential operational capacities that enable extensive clinical scale telemonitoring implementation.

VI. Result

A total of five refined clusters were developed during the clustering process through the addition of risk factors and demographic and mask and intervention data until the implementation of ANOVA-based feature importance created nine final cluster groups. The classification accuracy increased through improvement of treatment matching precision during the refinement step. Through cluster profiling the healthcare team generated unique treatment approaches to address specific needs among patients including fitting mask devices for leakage cases and providing behavior modification assistance to users with low use patterns. Survey reports together with intervention records enabled immediate cluster refinement and automatic treatment according to patients' shifting physiological traits and behavioral tendencies. The system brought improved medical oversight through risk score alerts that led medical staff to manage biometric tracking and patient care contact. HRV SpO_2 and ODI served as reliable biomarkers within the system framework because monitoring and therapeutic actions functioned effectively throughout different therapy phases. Different monitoring rates existed within the risk-specific protocol that required non-adherent patients to get daily evaluation but individuals with lower risk received evaluations only weekly or monthly. Patients' feedback demonstrated that the system successfully modified therapy interventions automatically which established the potential for large-scale AI-based useful management of CPAP adherence.

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